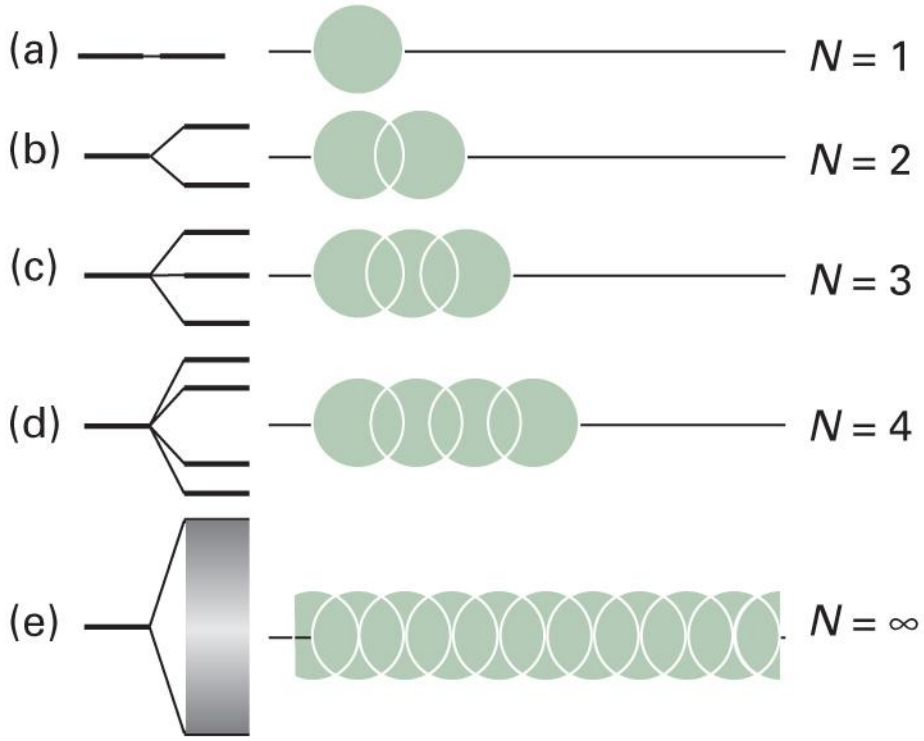
Physical Chemistry of Materials

Lecture 14

Magnetism

Last Class

Electronic Structures of Solids

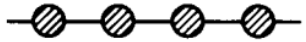


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 15C.6)

Bands

$$\psi_0 = \sum_n e^0 \chi_n = \sum_n \chi_n$$

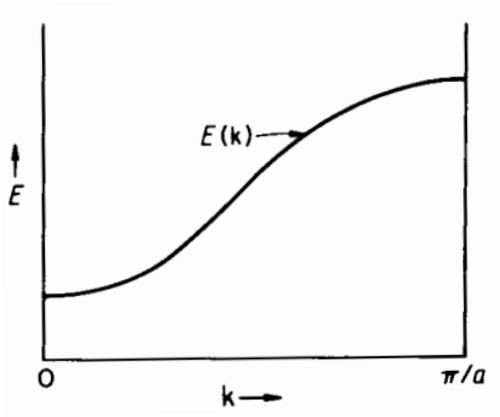
$$= \chi_0 + \chi_1 + \chi_2 + \chi_3 + \cdots$$



$$\psi_{\frac{\pi}{a}} = \sum_n e^{\pi i n} \chi_n = \sum_n (-1)^n \chi_n$$

$$= \chi_0 - \chi_1 + \chi_2 - \chi_3 + \cdots$$



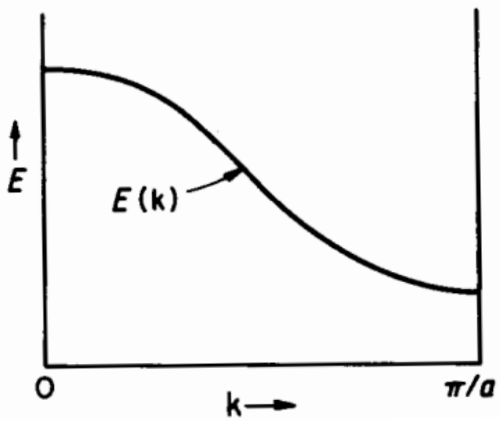


$$\psi_0 = \chi_0 + \chi_1 + \chi_2 + \chi_3 + \cdots$$



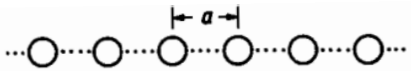
$$\psi_{\frac{\pi}{a}} = \chi_0 - \chi_1 + \chi_2 - \chi_3 + \cdots$$



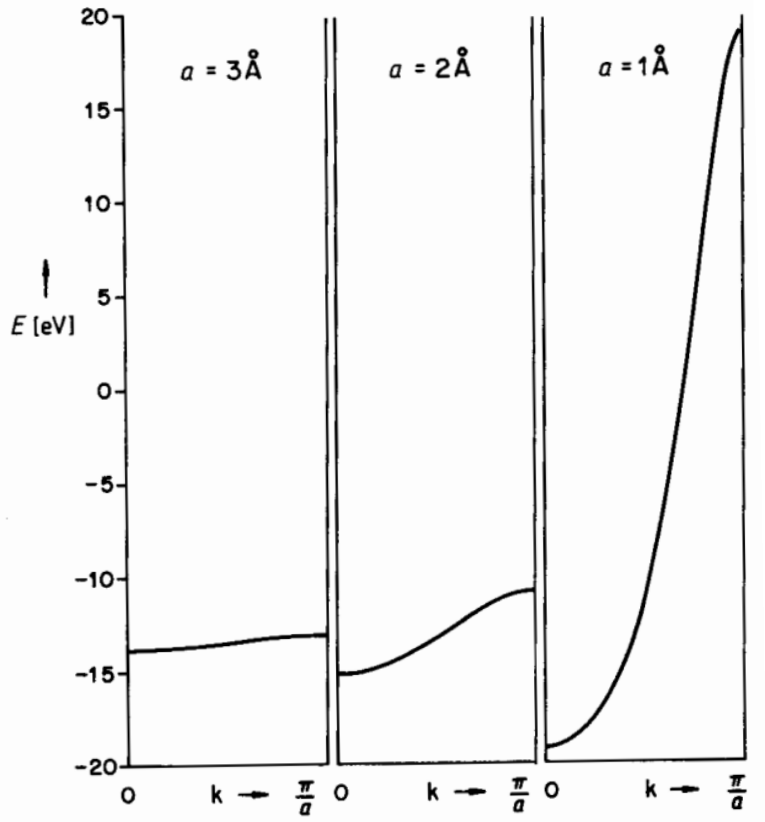


(Image: Hoffmann, *Solids and Surfaces*, p. 7 and p. 9)

"Band Width"

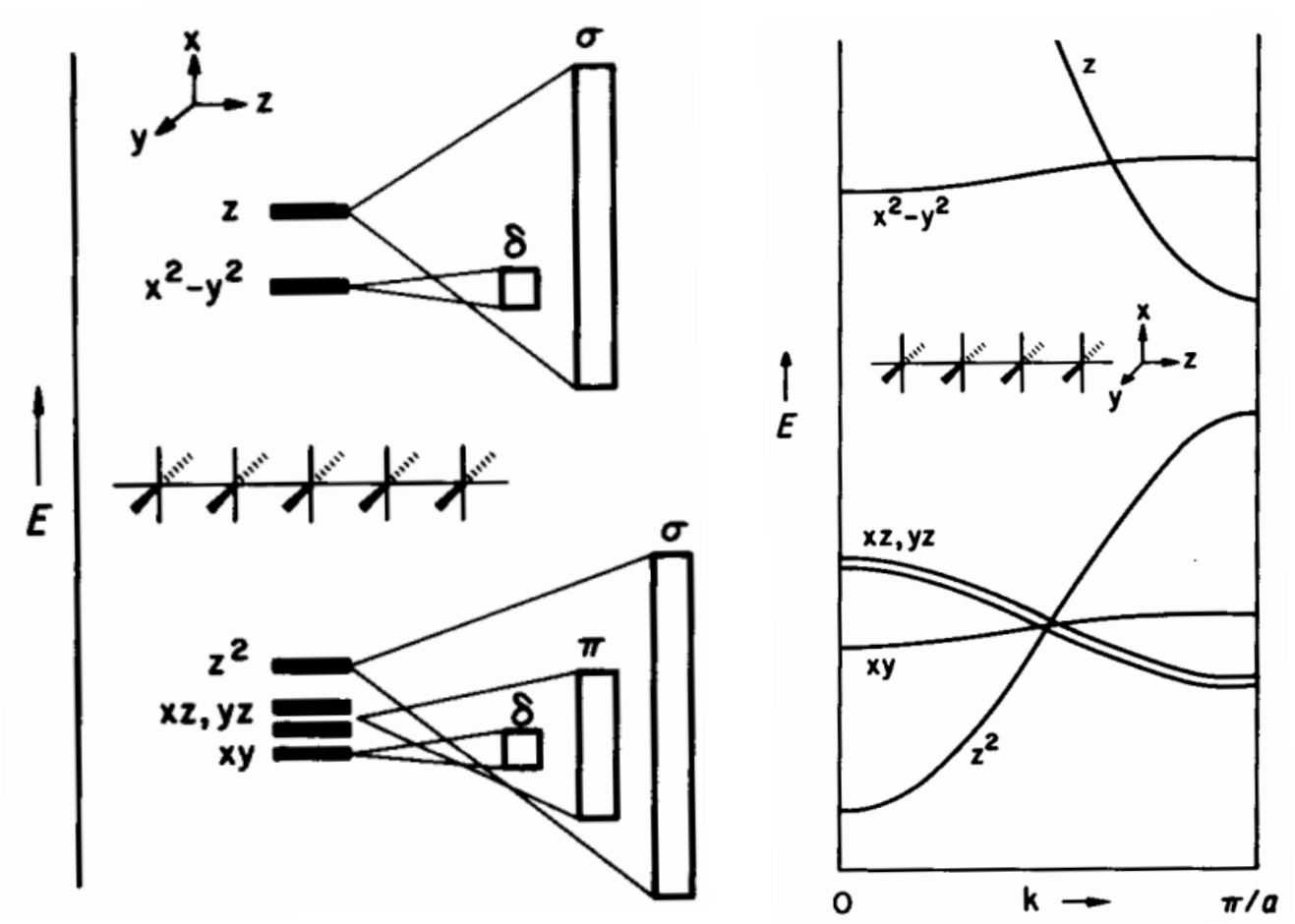


Chain of H atoms



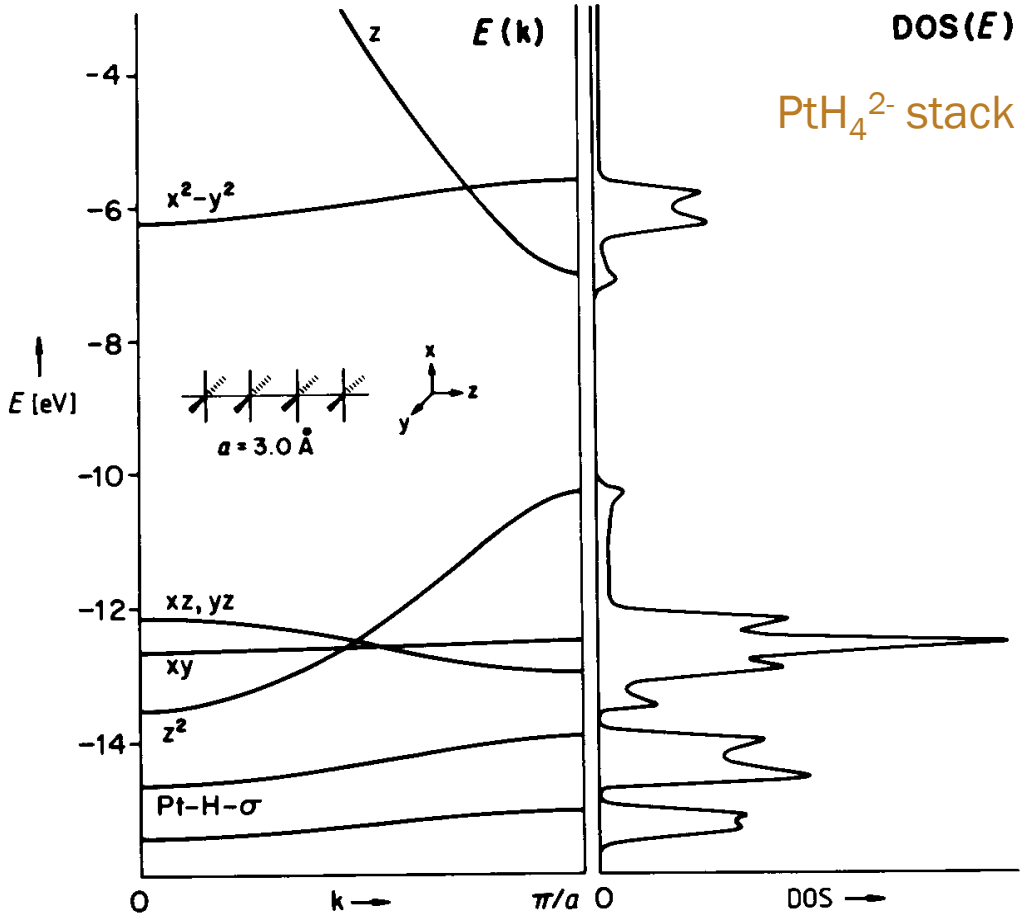
(Image: Hoffmann, *Solids and Surfaces*, Fig. 1)

Band Structure



(Image: Hoffmann, ***Solids and Surfaces***, p. 13)

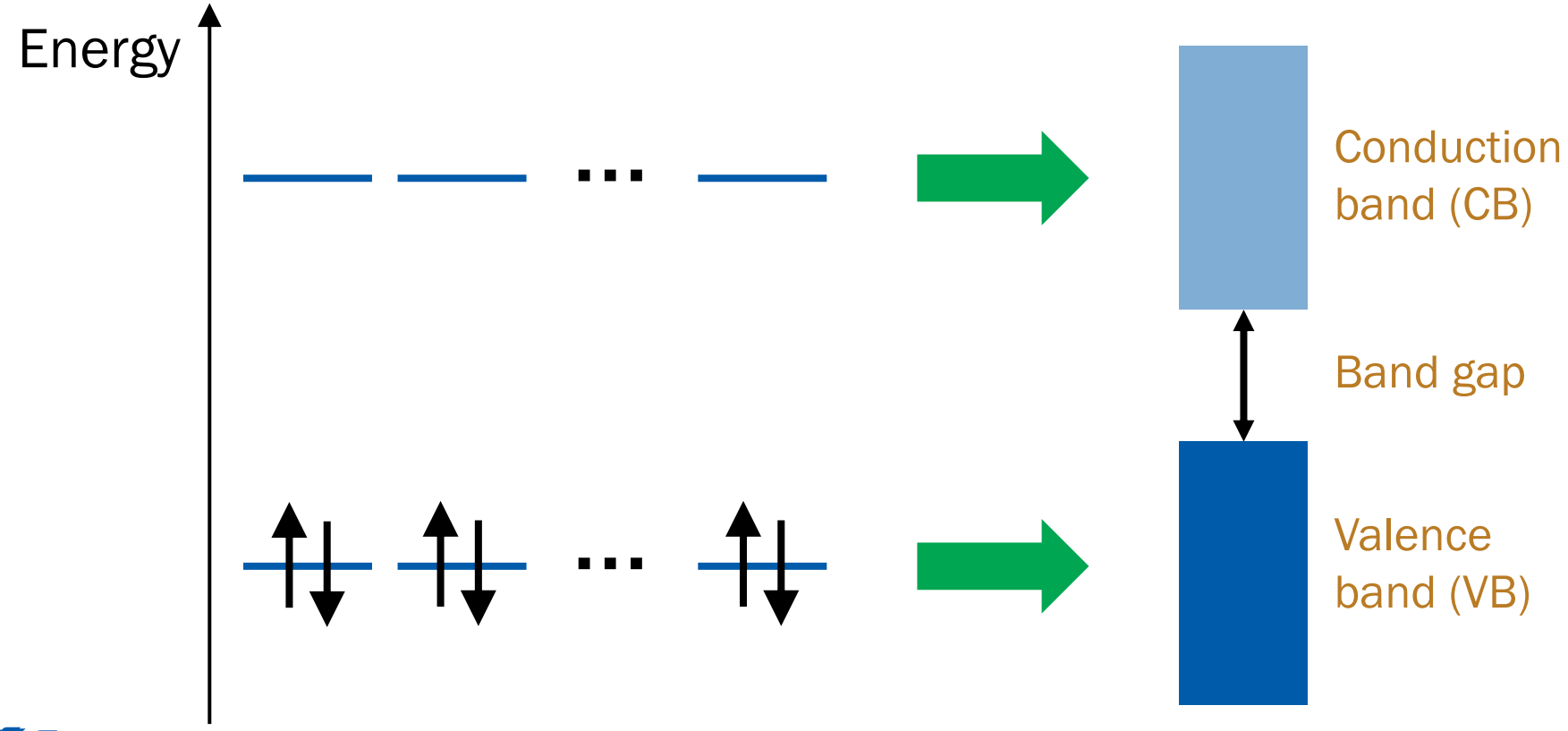
DOS (Density of States)



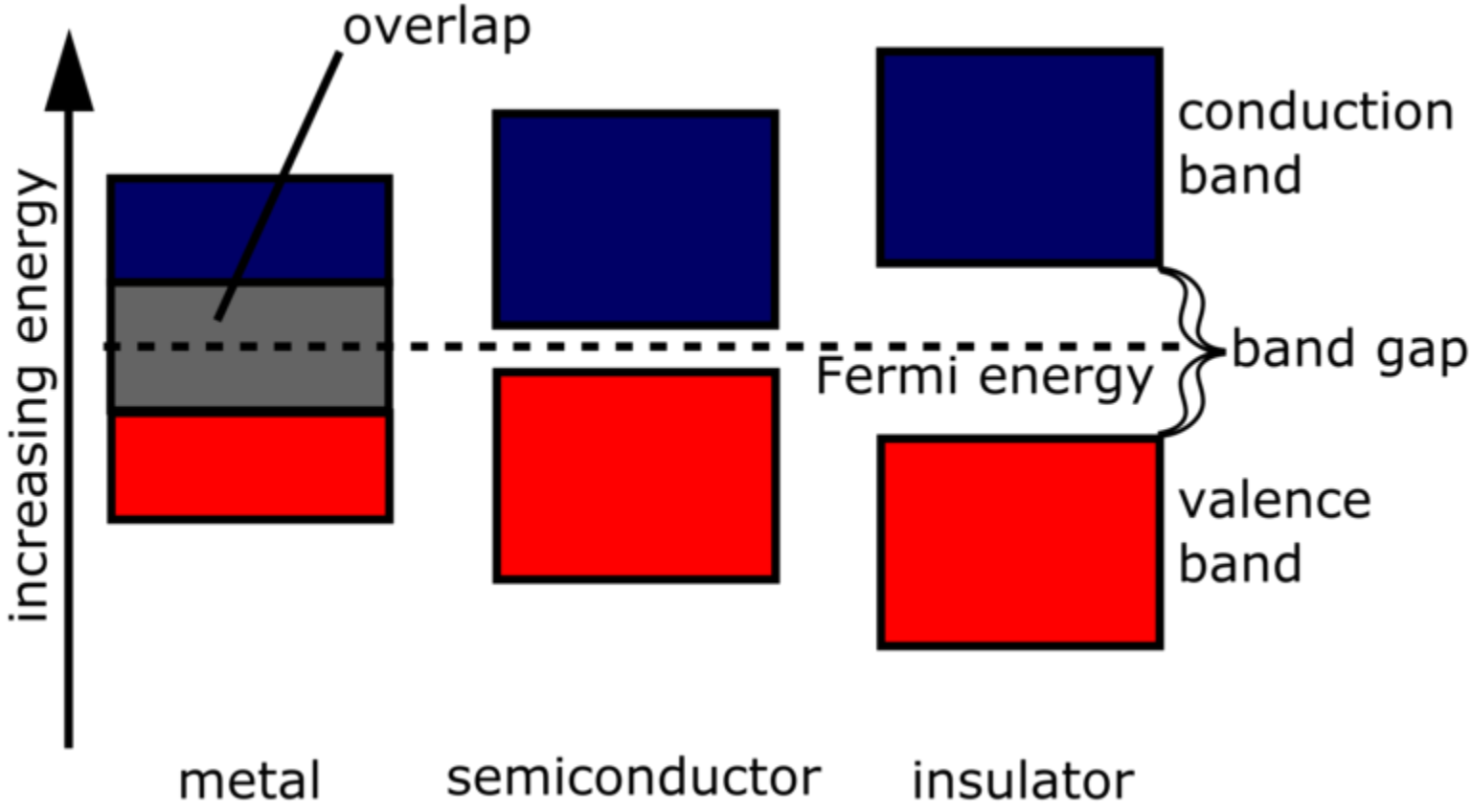
PtH_4^{2-} stack

(Image: Hoffmann, ***Solids and Surfaces***, p. 28)

Terminology

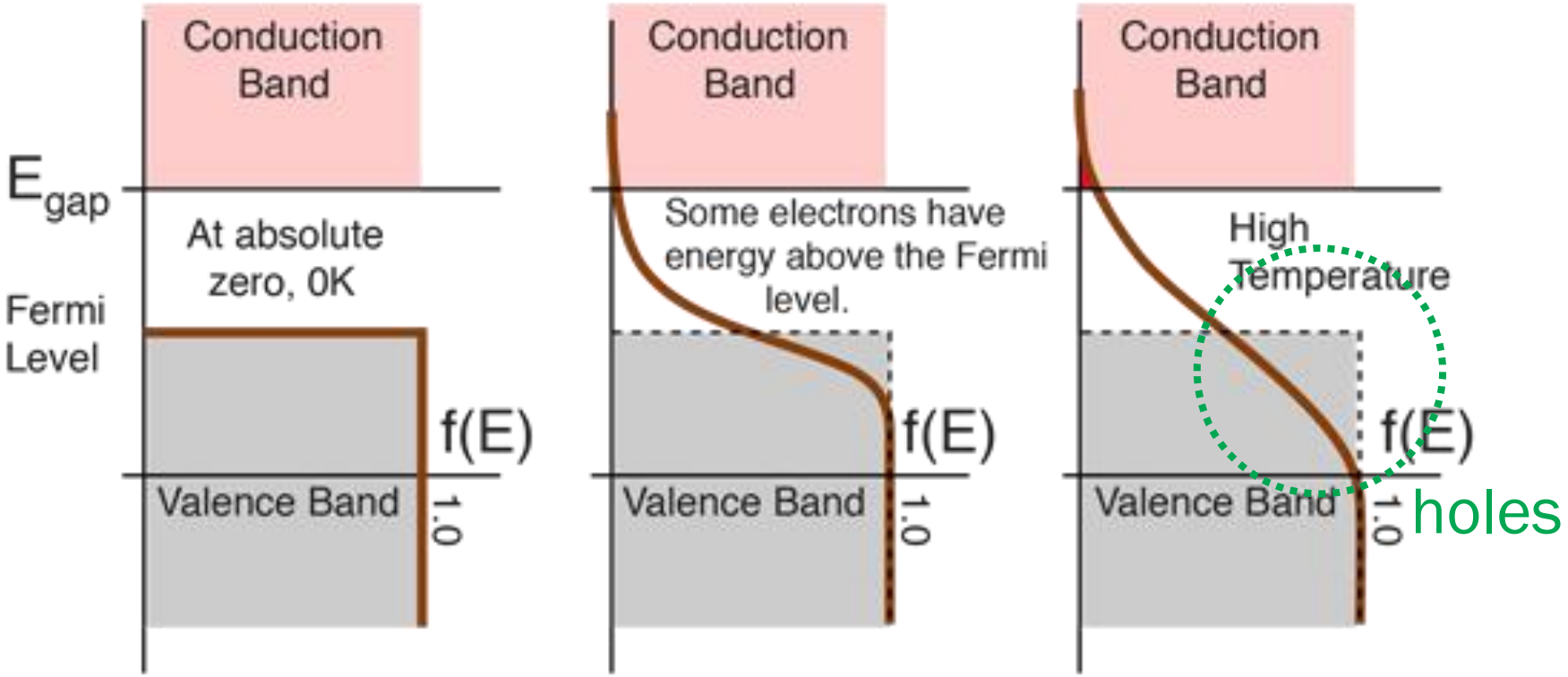


Band Gap and Conductivity



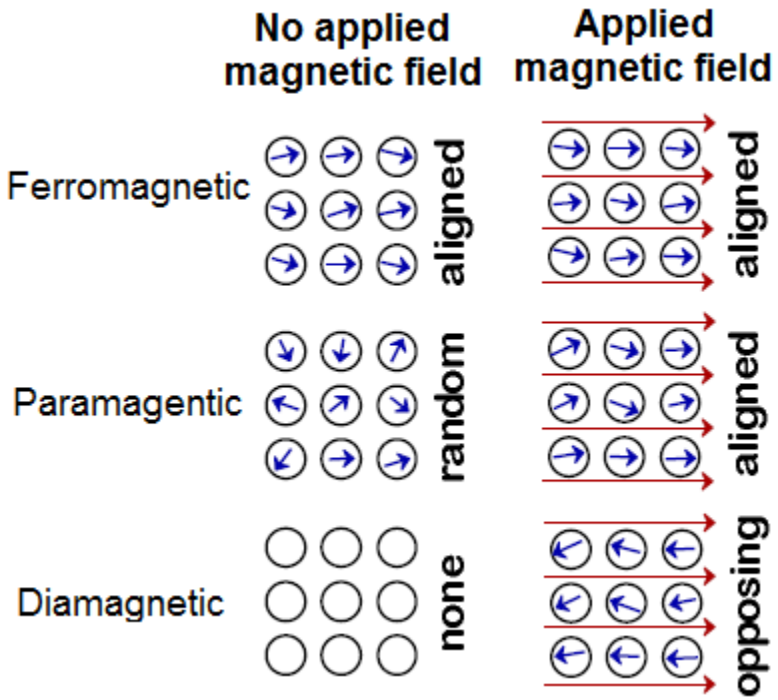
(Image: https://energyeducation.ca/encyclopedia/Band_gap)

At Finite Temperature



(Image: http://hyperphysics.phy-astr.gsu.edu/hbase/Solids/Fermi.html)

Magnetism



(Image: https://www.researchgate.net/publication/254705371_Understanding_the_Interaction_Between_Blood_Flow_and_an_Applied_Magnetic_Field)

Origin of Magnetization

For a free electron in a magnetic field,

$$E_{\text{spin}} = g\mu_{\text{B}} m_s B$$

where

- g: g-factor ($g \approx 2$)
- μ_{B}: Bohr magneton ($\mu_{\text{B}} = e\hbar/2m_{\text{e}}$)
- m_s: spin magnetic quantum number, $m_s = \pm 1/2$
- B: amplitude of the magnetic field

Microstates

$$E_{\text{spin}} = g\mu_{\text{B}} m_s B \approx 2\mu_{\text{B}} m_s B$$

Consider two spin states, up and down:

$$E(\uparrow) = \mu_{\text{B}} B$$

$$E(\downarrow) = -\mu_{\text{B}} B$$

With $\sigma = \pm 1$, we can also write

$$E(\sigma) = \mu_{\text{B}} B \sigma$$

Independent Spin Model

Consider a solid system consisting of N spins:



$$E(\uparrow) = +\mu_{\mathrm{B}} B, \quad E(\downarrow) = -\mu_{\mathrm{B}} B$$

The system consists of distinguishable particles:

$$Z = z^N$$

Partition Function

The one-particle partition function is

$$z = \sum_{\sigma} e^{-\beta E(\sigma)} = e^{-\beta E(\uparrow)} + e^{-\beta E(\downarrow)}$$

$$z = e^{-\beta \mu_{\mathrm{B}} B} + e^{+\beta \mu_{\mathrm{B}} B} = 2\cosh(\beta \mu_{\mathrm{B}} B)$$

Hence,

$$Z = z^N = [2\cosh(\beta \mu_{\mathrm{B}} B)]^N$$

Thermodynamics

- Helmholtz energy

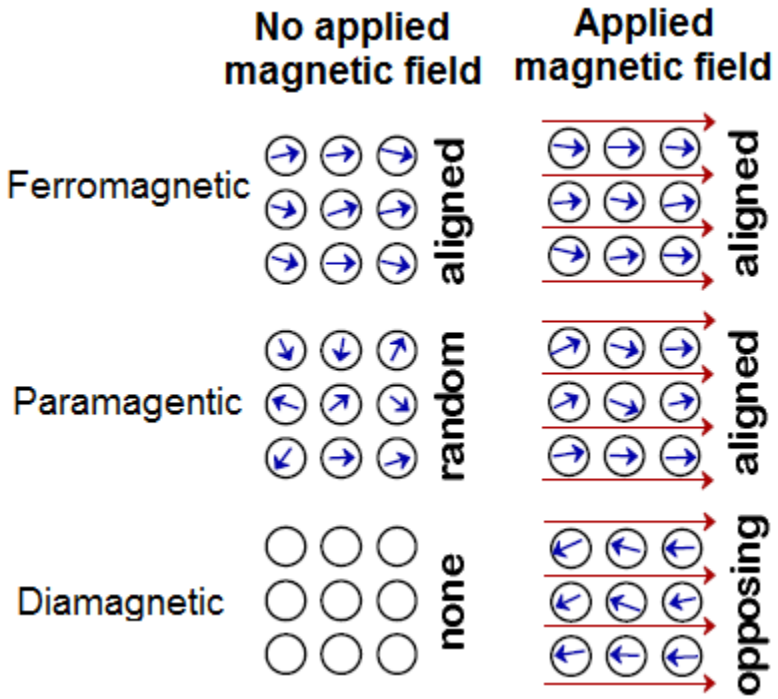
$$A = -k_{\mathrm{B}} T \ln Z = -N k_{\mathrm{B}} T \ln[2\cosh(\beta \mu_{\mathrm{B}} B)]$$

- Internal energy

$$U = -\frac{\partial \ln Z}{\partial \beta} = -N \frac{\partial}{\partial \beta} \ln[2\cosh(\beta \mu_{\mathrm{B}} B)]$$

$$\therefore \; U = -N \mu_{\mathrm{B}} B \tanh(\beta \mu_{\mathrm{B}} B)$$

Magnetism



$S \neq 0$ (or $J \neq 0$)

$S = 0$ (or $J = 0$)

(Image: https://www.researchgate.net/publication/254705371_Understanding_the_Interaction_Between_Blood_Flow_and_an_Applied_Magnetic_Field)

Ising Model

Consideration of neighboring spin-spin interactions

- System energy

$$E = \mu_{\mathrm{B}} B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

- Sign of J
  - $J > 0$: ferromagnetic
  - $J < 0$: antiferromagnetic

Partition Function of Ising Model

$$E = \mu_{\mathrm{B}} B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

$$Z = \sum_{\text{microstate}} e^{-\beta E} = \sum_{\{\sigma_i\}} e^{-\beta E(\{\sigma_i\})}$$

$$\sum_{\{\sigma_i\}} = \sum_{\sigma_1 = \pm 1} \sum_{\sigma_2 = \pm 1} \sum_{\sigma_3 = \pm 1} \cdots \sum_{\sigma_N = \pm 1}$$

Ising Model: Solved?

depends on the dimensionality D

- $D = 1$: exactly solvable for all J and $\mu_{\mathrm{B}} B$
- $D = 2$: exactly solvable for $\mu_{\mathrm{B}} B = 0$
- $D = 3$: not yet solved
- $D \geq 4$: solvable by the mean-field approach

Mean-Field Approach

$$E = \mu_{\mathrm{B}} B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

For spin i,

$$E_i = \mu_{\mathrm{B}} B \sigma_i - J \sigma_i \sum_{\mathrm{n.n.}j} \sigma_j \approx \mu_{\mathrm{B}} B \sigma_i - J \sigma_i \sum_{\mathrm{n.n.}j} \langle \sigma_j \rangle$$

By denoting $\langle \sigma_j \rangle = m$, for the coordination number z,

$$E_i = \mu_{\mathrm{B}} B \sigma_i - J \sigma_i z m$$

Mean-Field Approach

$$E_i = \mu_{\mathrm{B}} B \sigma_i - J \sigma_i z m = (\mu_{\mathrm{B}} B - J z m)\sigma_i$$

On the other hand,

$$m = \langle \sigma_i \rangle = \sum_{\sigma_i = \pm 1} \sigma_i p_i = \sum_{\sigma_i} \frac{\sigma_i e^{-\beta E_i}}{Z} = \frac{\sum_{\sigma_i} \sigma_i e^{-\beta E_i}}{\sum_{\sigma_i} e^{-\beta E_i}}$$

Thus,

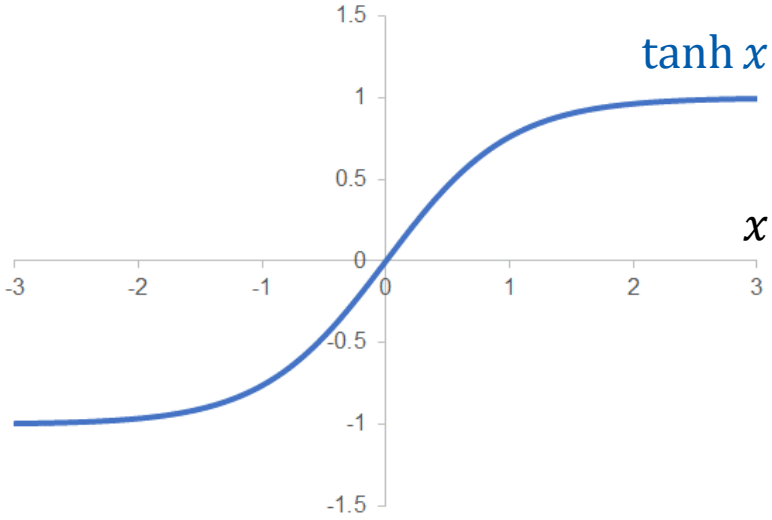
$$m = \frac{e^{-\beta(\mu_{\mathrm{B}} B - J z m)} - e^{\beta(\mu_{\mathrm{B}} B - J z m)}}{e^{-\beta(\mu_{\mathrm{B}} B - J z m)} + e^{\beta(\mu_{\mathrm{B}} B - J z m)}} = \tanh[\beta(J z m - \mu_{\mathrm{B}} B)]$$

(self-consistent equation)

Graphical Solution

$$m = \tanh[\beta(Jzm - \mu_B B)]$$

For $B = 0$, by letting $x = \beta Jzm$, we obtain $\frac{k_B T}{Jz} x = \tanh x$



Phase Transition

- High T (steep slope): one solution at $x = 0$
- Low T (shallow slope): three solutions
- Critical temperature T_c

$$\frac{k_B T}{Jz} x = \tanh x$$

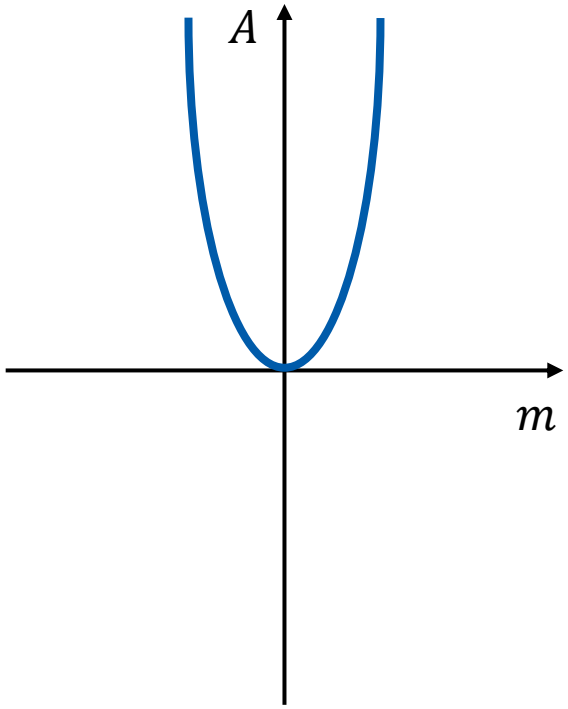
Since $\lim_{x \to 0} \tan x = x$,

$$k_B T_c = Jz$$

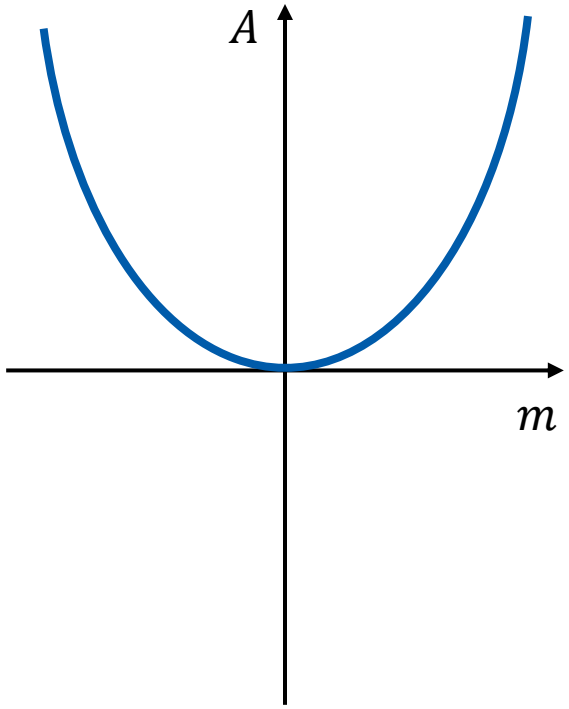
$$\therefore T_c = \frac{Jz}{k_B}$$

What Happens?

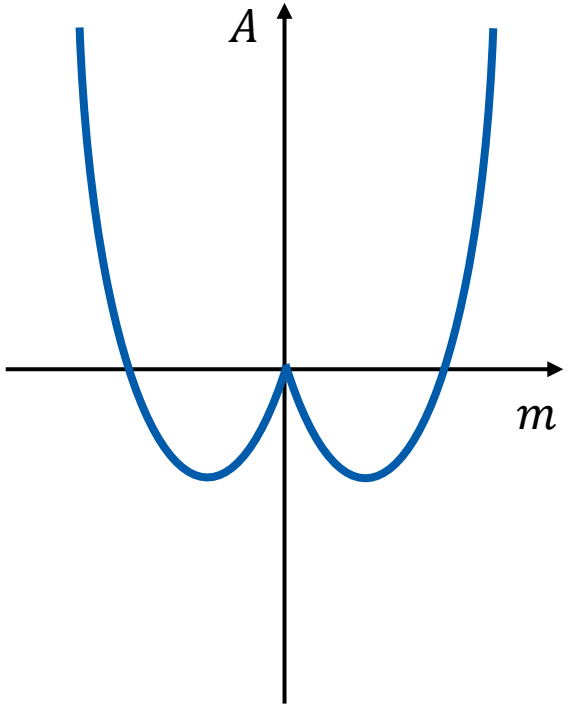
$T > T_c$



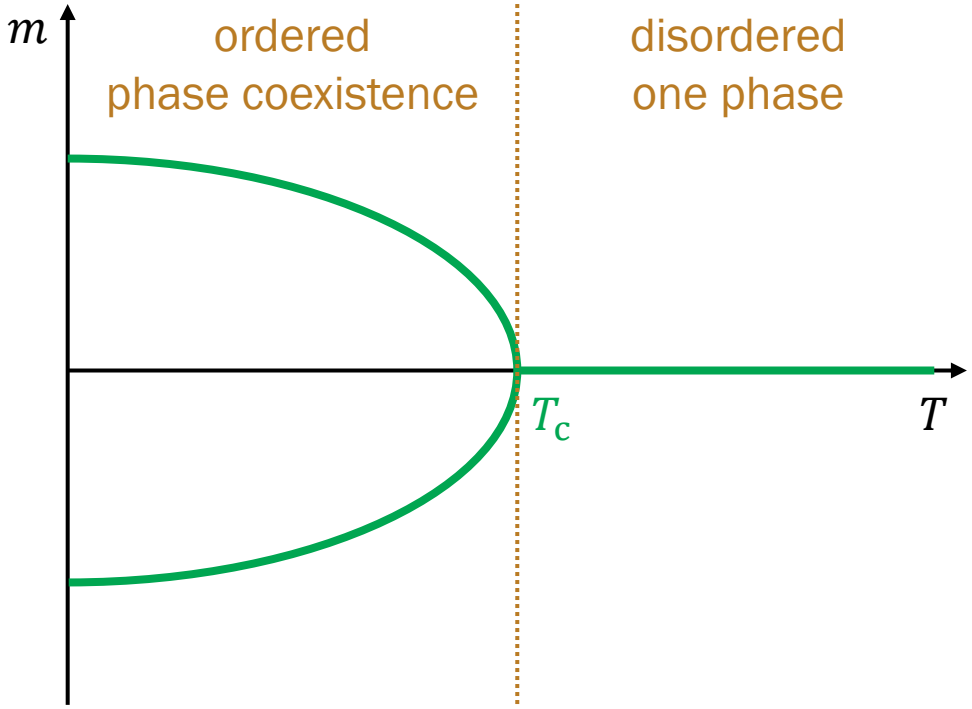
$T = T_c$



$T < T_c$



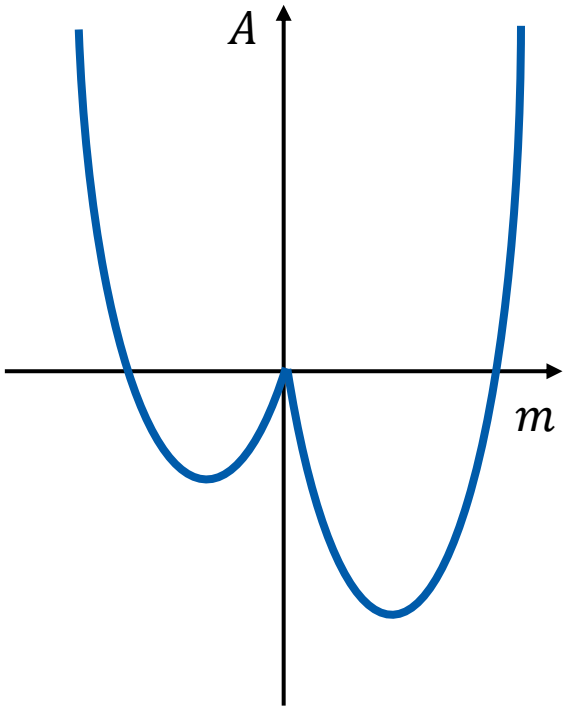
Positions of Minima



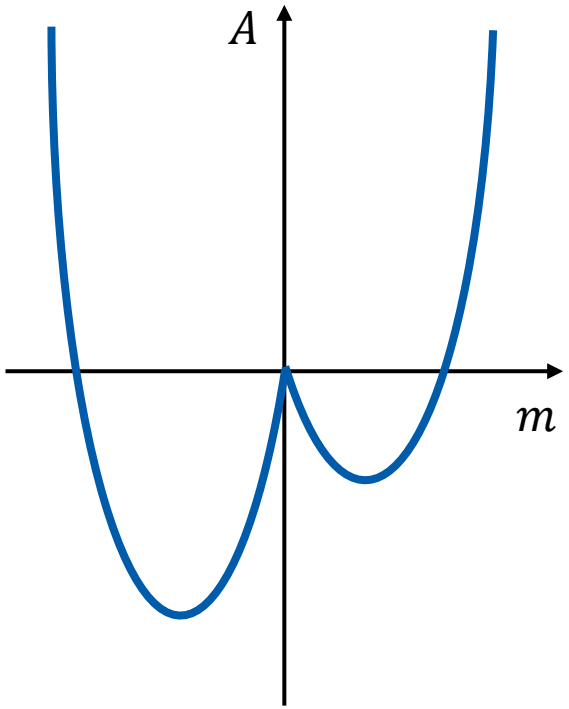
What If $B \neq 0$?

$(T < T_c)$

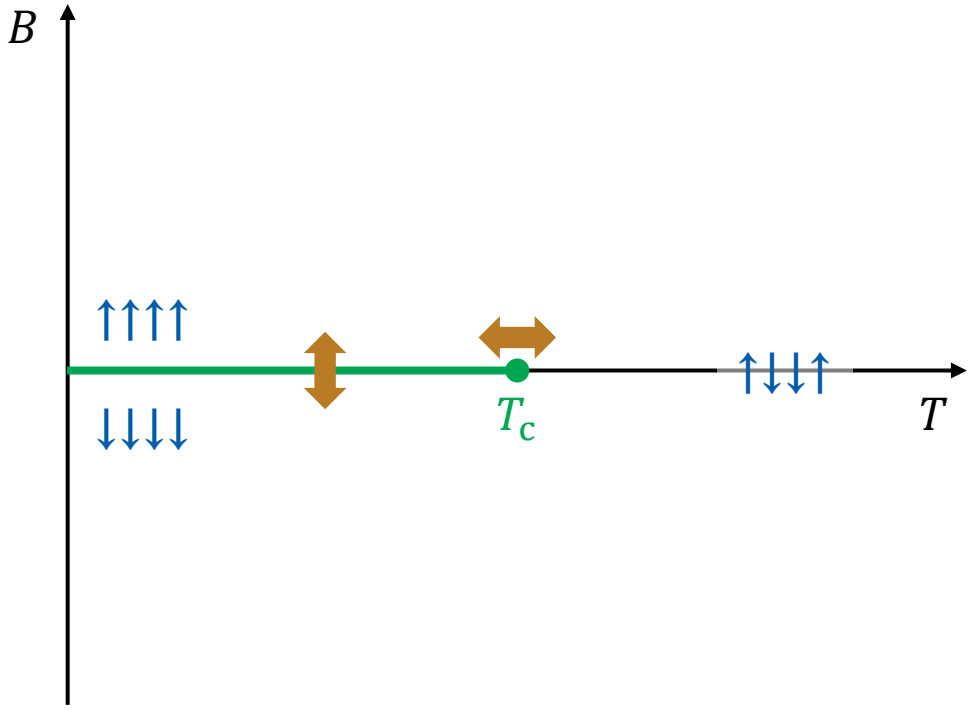
$B > 0$



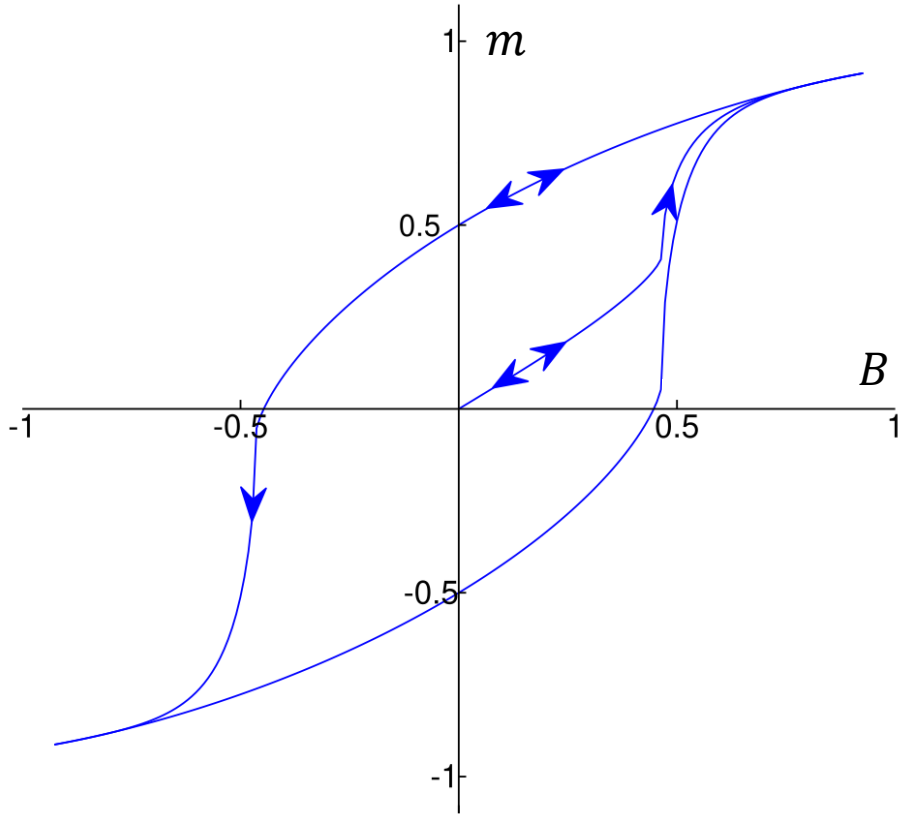
$B < 0$



Phase Diagram



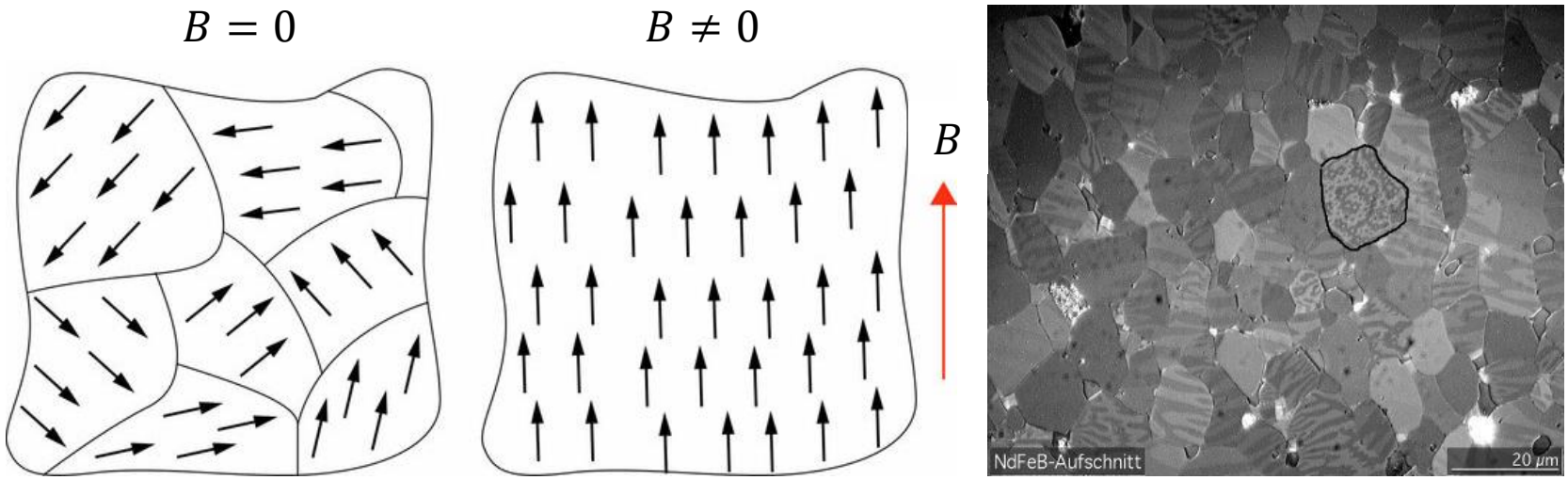
Magnetic Hysteresis



(Image: https://commons.wikimedia.org/wiki/File:StonerWohlfarthMainLoop.svg)

Magnetic Domains

Why is a ferromagnet not magnetic when $B = 0$?



(Images: https://i.sstatic.net/fKRdc.jpg and https://commons.wikimedia.org/wiki/File:NdFeB-Domains.jpg)

1-Dimensional Ising Model

For $B = 0$,

$$Z = \sum_{\{\sigma_i\}} e^{\beta J \sum_{\langle ij \rangle} \sigma_i \sigma_j}$$

Consider the periodic boundary condition ($\sigma_{N+1} = \sigma_1$);

$$Z = \sum_{\sigma_1 = \pm 1} \sum_{\sigma_2 = \pm 1} \sum_{\sigma_3 = \pm 1} \cdots \sum_{\sigma_N = \pm 1} e^{\beta J \sum_i \sigma_i \sigma_{i+1}}$$

1-Dimensional Ising Model

$$Z = \sum_{\sigma_1=\pm 1} \sum_{\sigma_2=\pm 1} \sum_{\sigma_3=\pm 1} \cdots \sum_{\sigma_N=\pm 1} e^{\beta J \sum_i \sigma_i \sigma_{i+1}}$$

Define $\tau_i = \sigma_i \sigma_{i+1} = \pm 1$;

$$Z = \sum_{\tau_1=\pm 1} \sum_{\tau_2=\pm 1} \sum_{\tau_3=\pm 1} \cdots \sum_{\tau_N=\pm 1} e^{\beta J \sum_i \tau_i}$$

$$Z = \left(\sum_{\tau=\pm 1} e^{\beta J \tau} \right)^N = [2\cosh(\beta J)]^N$$

Thermodynamics

- Partition function $Z = [2\cosh(\beta J)]^N$
- Helmholtz energy $A = -Nk_{\mathrm{B}}T\ln[2\cosh(\beta J)]$
- Energy $U = -NJ\tanh(\beta J)$

$$\lim_{T\to 0} U = -NJ$$

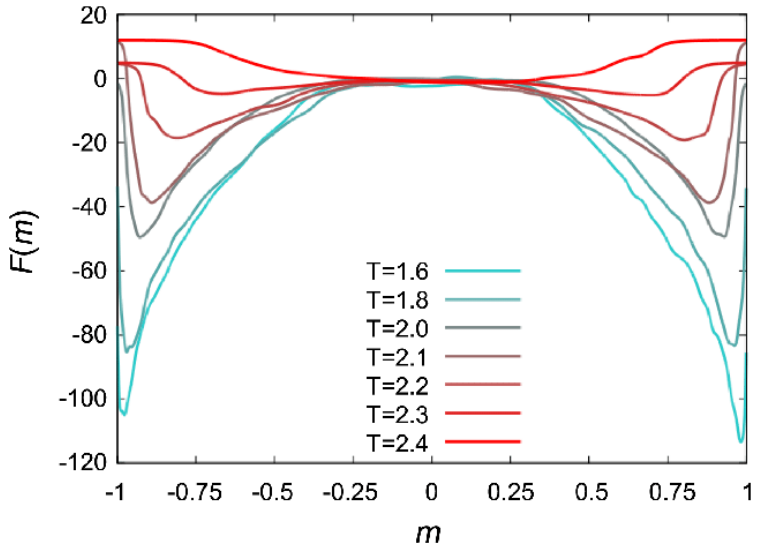
$$\lim_{T\to \infty} U = 0$$

No phase transition!

2D Ising Model

- Partition function (Onsager 1944)

$$\frac{\ln Z}{N} = \ln 2 + \frac{1}{8\pi^2}\int_{-\pi}^{\pi} \ln\left[\cosh^2(2K) - \sinh(2K)\left(\cos q_x + \cos q_y\right)\right] \mathrm{d}q_x \mathrm{d}q_y$$



(Image: Oh *et al.*, ***Bull Kor Chem Soc*** 2012)

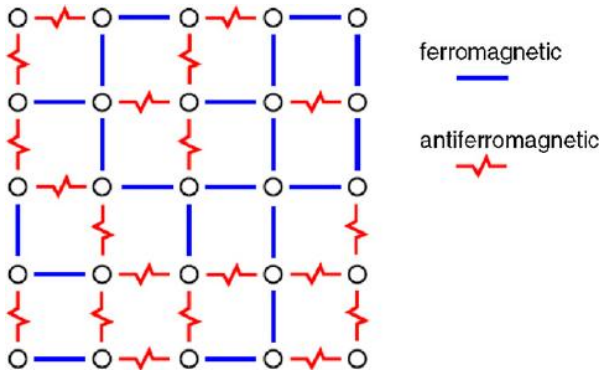
Spin Glasses

- Ising model

$$E = \mu_{\mathrm{B}} B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

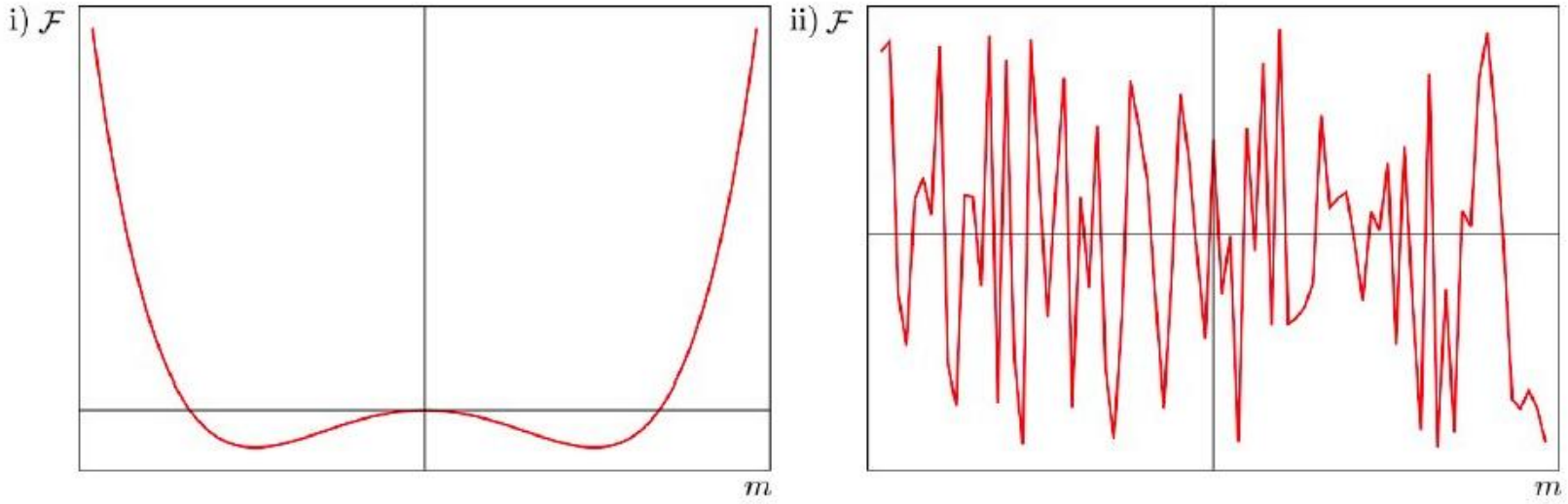
- Edwards-Anderson model

$$E = -\sum_{\langle ij \rangle} J_{ij} \sigma_i \sigma_j$$



(Image: https://www.researchgate.net/publication/280941761_Applied_Soft_Computing)

Disordered Systems





Nobel prize in Physics 2021

(Image: https://x.com/PhysicsSteve/status/1445348737157586946)

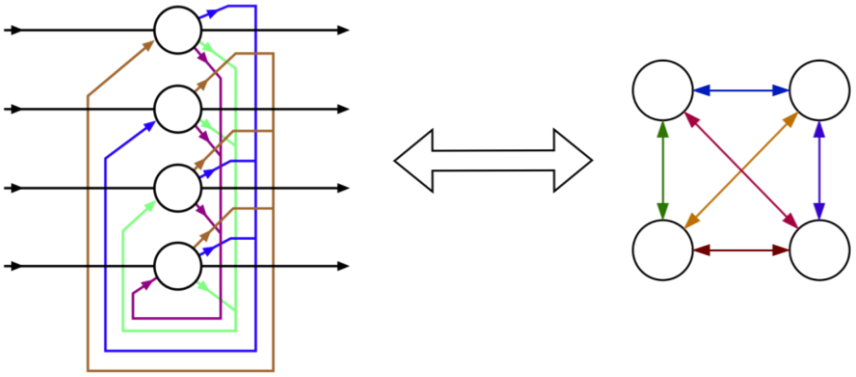
Hopfield Model

- Time evolution

$$\sigma_i(t + \Delta t) = \mathrm{sgn}\left(\sum_j J_{ij}\sigma_j(t)\right)$$

- Equivalent energy

$$E = -\sum_{\langle ij \rangle} J_{ij}\sigma_i\sigma_j$$





Nobel prize in Physics 2024

(Image: https://www.zach.be/p/implementing-hopfield-networks-on)